Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details:

Batch Number: 1

Start Date:13-Feb -

2026

Names:

Sr.No.	Name
1	Nagula Rachana
2	Gopikashree
3	Abhilash Kumar Jha
4	Shiven Pratap Singh
5	Gyan Anand
6	Riddhi Hajare

Internship Duration: 10 Weeks

1. Project Title

HotelRevAI AI-Driven Revenue Analysis for Hotels

2. Project Objective

The primary objective of the **HotelRevAI – AI-Driven Revenue Analysis for Hotels** project is to design and implement an intelligent analytics system that enables hotels to monitor, analyze, and optimize their revenue and occupancy performance using data-driven insights.

The specific objectives of the project are as follows:

- To collect, integrate, and structure hotel booking, customer, and room-level data into a well-defined analytical data model suitable for reporting and analysis.
- To calculate and analyze key hospitality performance indicators such as **Occupancy Rate, Average Daily Rate (ADR), and Revenue per Available Room (RevPAR)** across different time periods.
- To study guest behavior patterns by analyzing booking sources, guest types, nationality, length of stay, and spending behavior.
- To identify customer segments such as first-time guests, repeat customers, and high-value guests using clustering techniques.
- To analyze booking trends, cancellation behavior, lead time patterns, and no-show rates to support operational planning.
- To forecast future occupancy trends and demand patterns to assist hotel management in proactive decision-making.
- To develop an interactive, role-based dashboard that provides actionable revenue strategy insights for General Managers (GMs) and Revenue Managers (RMs).

3. Project Description in Detail

HotelRevAI is an AI-driven revenue analytics solution designed to help hotels make informed strategic decisions by transforming raw booking and customer data into meaningful business insights. The system integrates multiple datasets including booking details, customer information, room attributes, and hotel branch data to build a unified analytical foundation.

The project begins with data ingestion and modeling, where raw data is cleaned, transformed, and organized into a **star schema** consisting of fact tables and dimension tables such as Date, Room, Customer, and Hotel Branch. Derived metrics such as booking duration, stay type, and room category are calculated to enrich the dataset.

Using this structured data, the system computes essential hotel performance metrics including **Occupancy Percentage, ADR, and RevPAR**, enabling analysis of daily, weekly, monthly, and seasonal performance. Comparative analysis between **direct bookings and Online Travel Agency (OTA) bookings** helps assess distribution channel effectiveness.

The Guest Analysis module focuses on understanding customer behavior by analyzing guest demographics, booking patterns, nationality distribution, and stay duration. Customers are further segmented into meaningful clusters such as business travellers, family guests, loyal customers, and high spenders to support targeted marketing and personalization strategies.

The Forecasting and Cancellation module analyzes historical trends to predict future occupancy levels, identify high-risk cancellation periods, and study lead time distributions. Visual insights into no-show trends and refund patterns help hotels reduce revenue leakage and improve forecasting accuracy.

Finally, the **Revenue Strategy Dashboard** consolidates all insights into an interactive Power BI dashboard designed for hotel decision-makers. The dashboard highlights upselling opportunities, seasonal pricing recommendations, and room-type-based revenue strategies, enabling hotel management to improve profitability and operational efficiency.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project kickoff, understanding hotel revenue management concepts, and identifying data requirements.	Conducted project initiation, finalized project scope, identified key KPIs such as Occupancy %, ADR, and RevPAR, and reviewed hotel datasets.
Week 2	Data collection and initial data modeling for booking, customer, and room datasets.	Collected booking, customer, room, and hotel branch data; performed initial data cleaning and validation.
Week 3	Designing analytical data model and building star schema.	Designed and implemented a star schema with Fact Bookings and Dimension tables (Date, Room, Customer, Hotel Branch).
Week 4	KPI calculation and occupancy & revenue analysis.	Calculated Occupancy %, ADR, and RevPAR; developed daily, weekly, and seasonal performance visuals.
Week 5	Guest analysis and demographic segmentation planning.	Analyzed guest types (business, family, solo), nationality distribution, booking sources, and stay duration patterns.
Week 6	Customer segmentation and behavioral analysis.	Clustered customers into first-time guests, loyal customers, and high spenders; validated segmentation insights.
Week 7	Forecasting and cancellation trend analysis.	Analyzed booking trends, cancellation rates, lead time distribution, and no-show patterns; created trend-based forecasts.
Week 8	Advanced forecasting and trend visualization.	Refined occupancy trend analysis and integrated cancellation and refund visuals into the dashboard.
Week 9	Revenue strategy identification and dashboard integration.	Identified upselling opportunities, seasonal pricing strategies, and room-type-based revenue insights.
Week 10	Final dashboard completion, documentation, and submission.	Completed end-to-end interactive Power BI dashboard and finalized project documentation and presentation.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Initial project briefing, finalization of objectives, understanding hotel revenue analytics concepts, and identification of datasets and KPIs.	Week 1
Data Model Completion	Completion of data cleaning, transformation, and implementation of star schema with fact and dimension tables.	Week 3
KPI Dashboard Prototype	Development of core occupancy and revenue metrics including Occupancy %, ADR, and RevPAR with time-based analysis.	Week 4
Guest Analysis Module Review	Completion of guest demographic analysis, booking source insights, and customer segmentation into meaningful clusters.	Week 6
Forecasting & Trend Analysis Completion	Implementation of booking trend analysis, cancellation behavior, lead time distribution, and occupancy forecasting visuals.	Week 8
Final Submission & Presentation	Full integration of all modules into a single interactive dashboard along with finalized documentation and presentation.	Week 10

5b. Project Execution Details

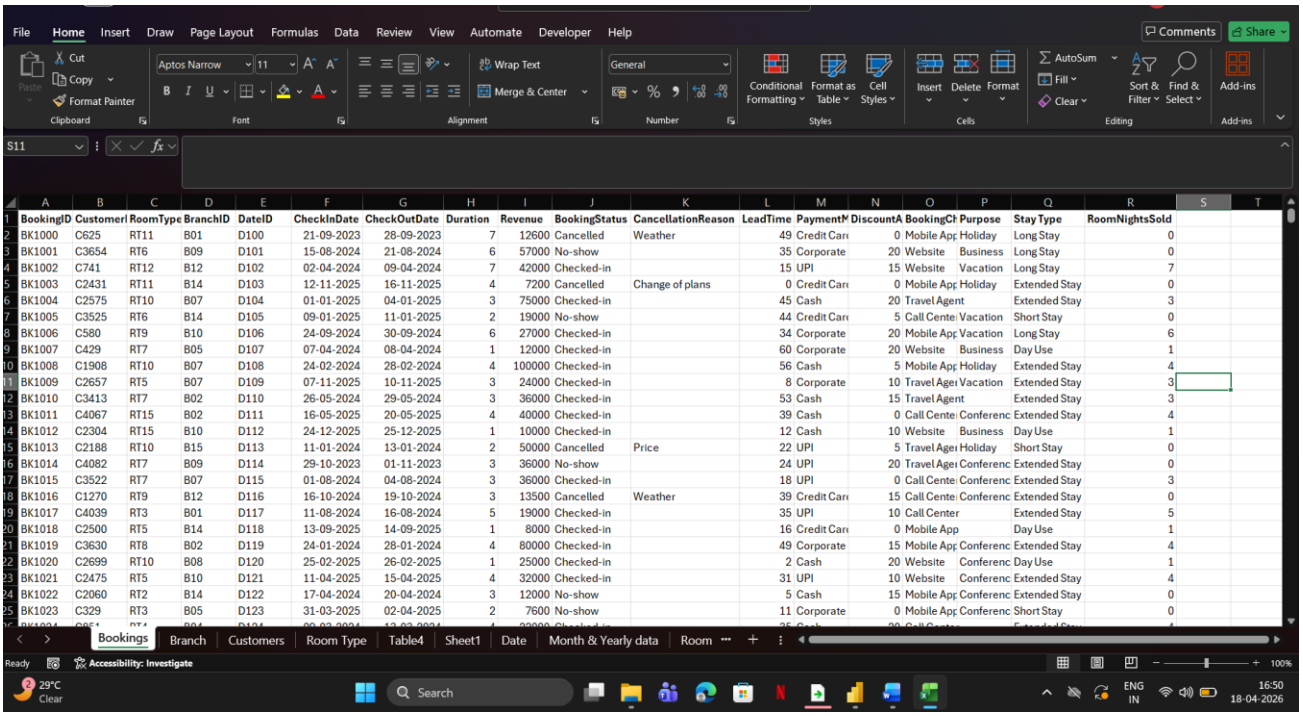
The execution of the **HotelRevAI** project followed a modular and structured approach. Initial phases focused on data understanding, cleaning, and transformation using analytical best practices. The data model was designed to ensure scalability and efficient reporting.

Key performance metrics were calculated using business logic aligned with hospitality industry standards. Visual analytics were created using interactive charts, slicers, and drill-down features to allow dynamic exploration of data. Forecasting trends were derived using historical patterns and trend lines to support future planning.

The final dashboard integrates operational, analytical, and strategic views, enabling hotel stakeholders to monitor performance, identify revenue opportunities, and optimize pricing and distribution strategies effectively.

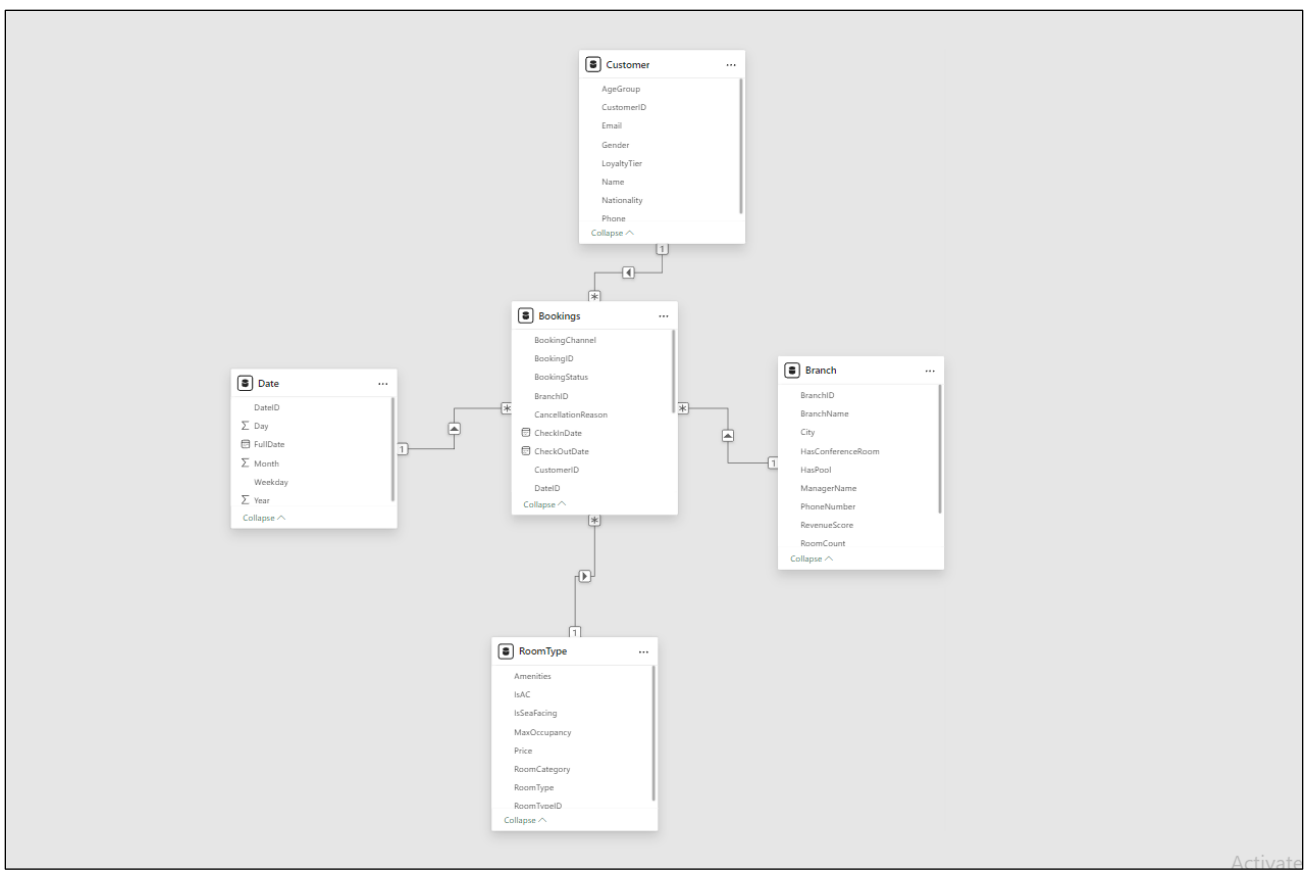
6. Snapshots / Screenshots

1. Dataset



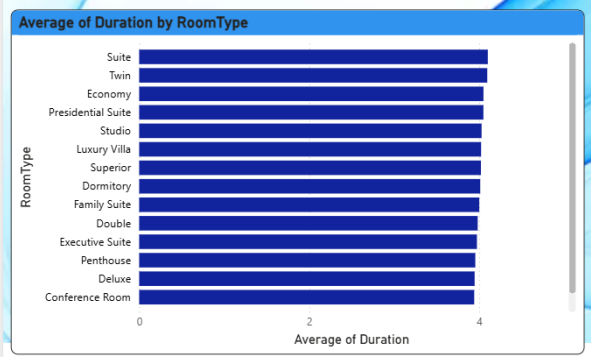
BookingID	CustomerI	RoomType	BranchID	DateID	CheckInDate	CheckOutDate	Duration	Revenue	BookingStatus	CancellationReason	LeadTime	PaymentM	DiscountA	BookingCh	Purpose	Stay Type	RoomNightsSold
BK1000	C625	RT11	B01	D100	21-09-2023	28-09-2023	7	12600	Cancelled	Weather	49	Credit Car	0	Mobile App	Holiday	Long Stay	0
BK1001	C3654	RT6	B09	D101	15-08-2024	21-08-2024	6	57000	No-show		35	Corporate	20	Website	Business	Long Stay	0
BK1002	C741	RT12	B12	D102	02-04-2024	09-04-2024	7	42000	Checked-in		15	UPI	15	Website	Vacation	Long Stay	7
BK1003	C2431	RT11	B14	D103	12-11-2025	16-11-2025	4	7200	Cancelled	Change of plans	0	Credit Car	0	Mobile App	Holiday	Extended Stay	0
BK1004	C2575	RT10	B07	D104	01-01-2025	04-01-2025	3	75000	Checked-in		45	Cash	20	Travel Agent		Extended Stay	3
BK1005	C3525	RT6	B14	D105	09-01-2025	11-01-2025	2	19000	No-show		44	Credit Car	5	Call Cente	Vacation	Short Stay	0
BK1006	C580	RT9	B10	D106	24-09-2024	30-09-2024	6	27000	Checked-in		34	Corporate	20	Mobile App	Vacation	Long Stay	6
BK1007	C429	RT7	B05	D107	07-04-2024	08-04-2024	1	12000	Checked-in		60	Corporate	20	Website	Business	Day Use	1
BK1008	C1908	RT10	B07	D108	24-02-2024	28-02-2024	4	100000	Checked-in		56	Cash	5	Mobile App	Holiday	Extended Stay	4
BK1009	C2657	RT5	B07	D109	07-11-2025	10-11-2025	3	24000	Checked-in		8	Corporate	10	Travel Agen	Vacation	Extended Stay	3
BK1010	C3413	RT7	B02	D110	26-05-2024	29-05-2024	3	36000	Checked-in		53	Cash	15	Travel Agent		Extended Stay	3
BK1011	C4067	RT15	B02	D111	16-05-2025	20-05-2025	4	40000	Checked-in		39	Cash	0	Call Cente	Conferenc	Extended Stay	4
BK1012	C2304	RT15	B10	D112	24-12-2025	25-12-2025	1	10000	Checked-in		12	Cash	10	Website	Business	Day Use	1
BK1013	C2188	RT10	B15	D113	11-01-2024	13-01-2024	2	50000	Cancelled	Price	22	UPI	5	Travel Agen	Holiday	Short Stay	0
BK1014	C4082	RT7	B09	D114	29-10-2023	01-11-2023	3	36000	No-show		24	UPI	20	Travel Agen	Conferenc	Extended Stay	0
BK1015	C3522	RT7	B07	D115	01-08-2024	04-08-2024	3	36000	Checked-in		18	UPI	0	Call Cente	Conferenc	Extended Stay	3
BK1016	C1270	RT9	B12	D116	16-10-2024	19-10-2024	3	13500	Cancelled	Weather	39	Credit Car	15	Call Cente	Conferenc	Extended Stay	0
BK1017	C4039	RT3	B01	D117	11-08-2024	16-08-2024	5	19000	Checked-in		35	UPI	10	Call Center		Extended Stay	5
BK1018	C2500	RT5	B14	D118	13-09-2025	14-09-2025	1	8000	Checked-in		16	Credit Car	0	Mobile App		Day Use	1
BK1019	C3630	RT8	B02	D119	24-01-2024	28-01-2024	4	80000	Checked-in		49	Corporate	15	Mobile App	Conferenc	Extended Stay	4
BK1020	C2699	RT10	B08	D120	25-02-2025	26-02-2025	1	25000	Checked-in		2	Cash	20	Website	Conferenc	Day Use	1
BK1021	C2475	RT5	B10	D121	11-04-2025	15-04-2025	4	32000	Checked-in		31	UPI	10	Website	Conferenc	Extended Stay	4
BK1022	C2060	RT2	B14	D122	17-04-2024	20-04-2024	3	12000	No-show		5	Cash	15	Mobile App	Conferenc	Extended Stay	0
BK1023	C329	RT3	B05	D123	31-03-2025	02-04-2025	2	7600	No-show		11	Corporate	0	Mobile App	Conferenc	Short Stay	0

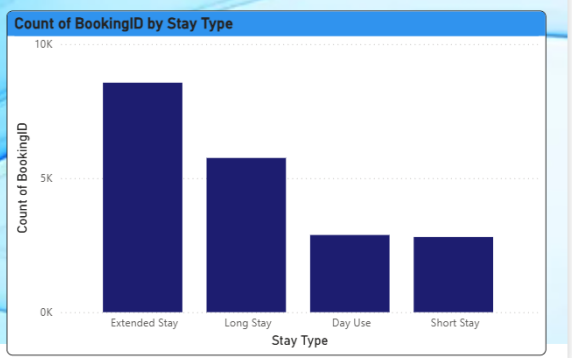
2. Star Schema



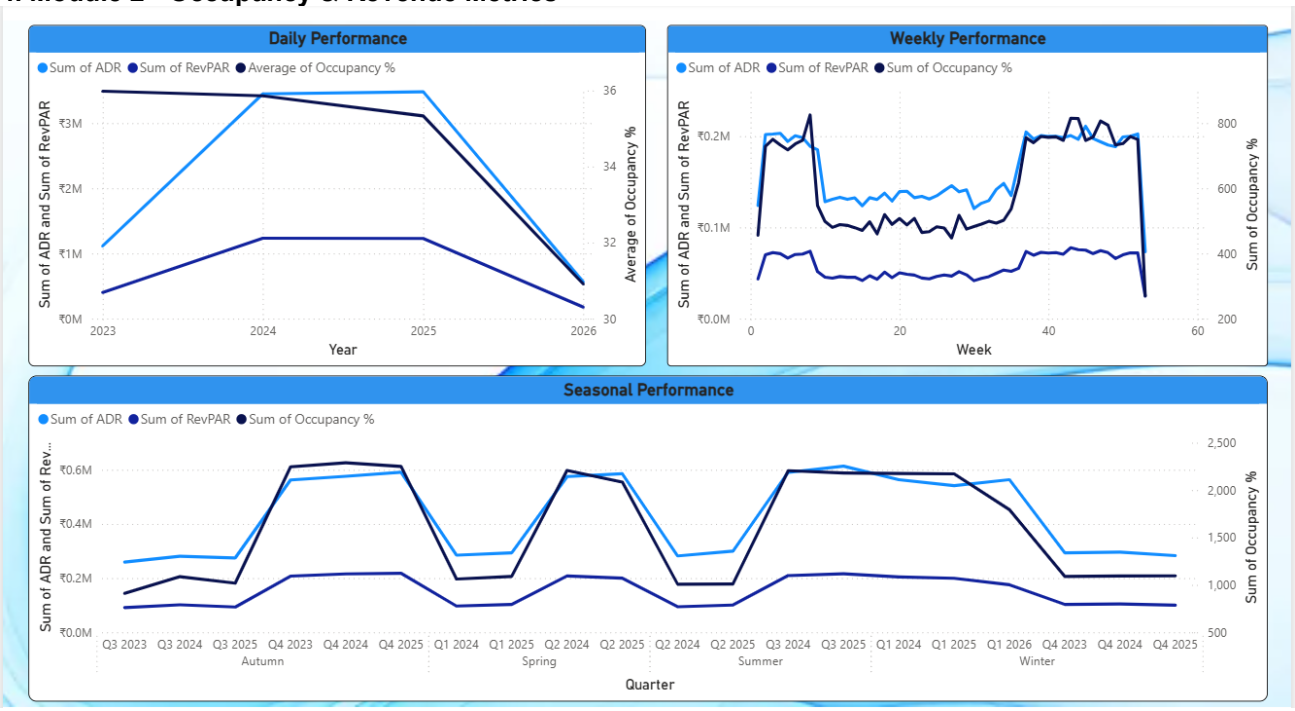
3. Module 1 - Data Modeling and Ingestion

BranchName	Conference Room	Deluxe	Dormitory	Double	Economy	Executive Suite	Family Suite	Luxury Villa	Penthouse	Presidential Suite	Single	Studio	Suite	Superior
ITC Grand Bharat	90	79	81	98	90	87	88	86	75	86	89	88	91	89
ITC Grand Central	86	99	91	112	93	96	69	81	87	91	99	85	88	76
ITC Grand Chola	90	100	106	88	102	98	92	79	66	96	89	91	92	111
ITC Grand Goa Resort & Spa	109	98	94	100	89	82	95	88	77	85	89	94	102	93
ITC Kakatiya	95	96	102	93	86	88	97	97	81	86	86	89	90	84
ITC Kohenur	88	94	85	94	88	83	97	98	99	117	67	102	92	97
ITC Maratha	80	97	99	87	80	92	81	76	65	84	99	88	88	94
ITC Maurya	83	105	95	79	102	87	84	82	92	93	86	82	86	74
ITC Mughal	79	92	85	81	91	84	84	92	93	101	76	80	92	71
ITC Narmada	82	98	93	89	97	66	87	102	76	91	81	96	98	94
Total	**1333**	**1400**	**1369**	**1364**	**1371**	**1282**	**1303**	**1343**	**1263**	**1326**	**1308**	**1319**	**1363**	**1328**



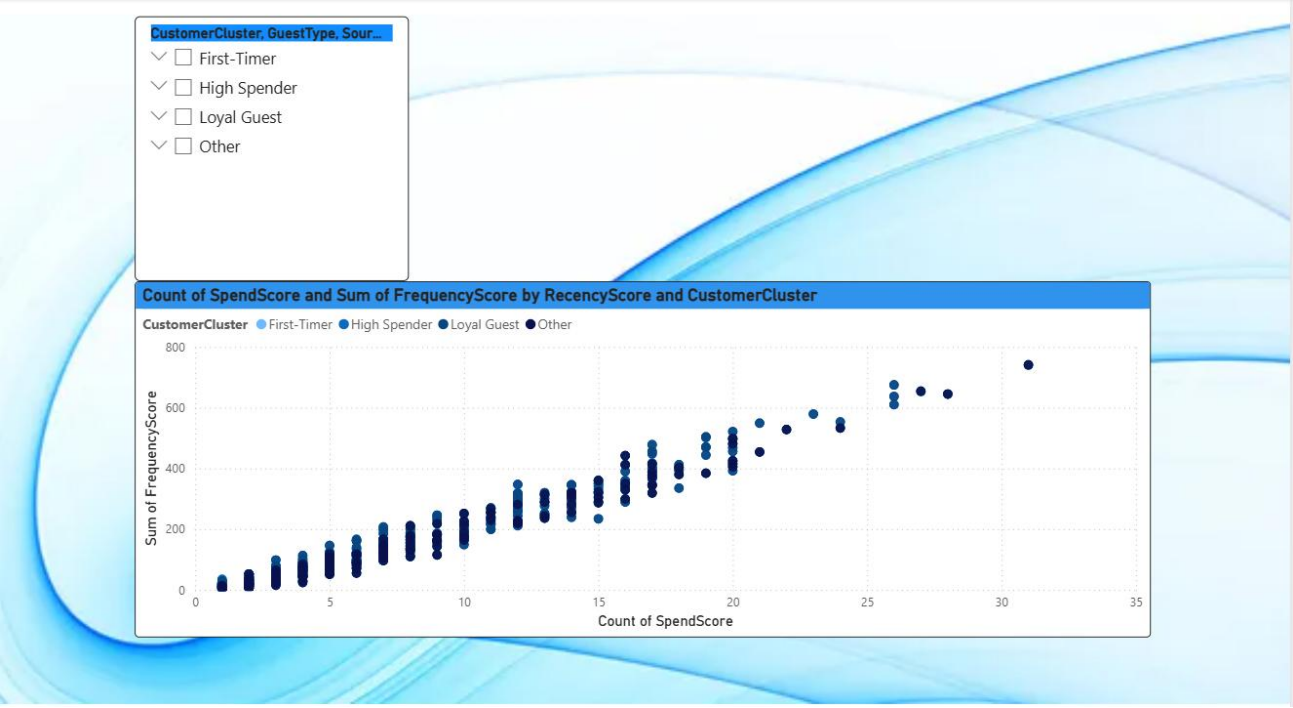


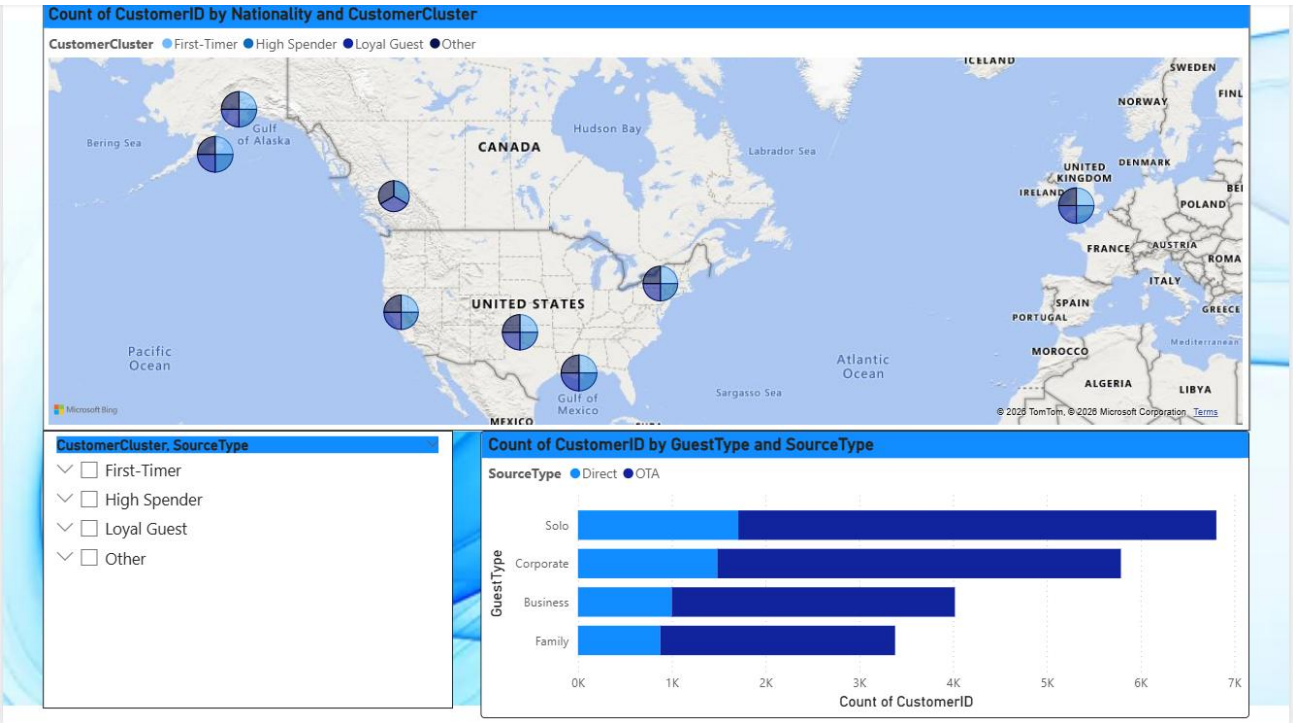
4. Module 2 - Occupancy & Revenue Metrics



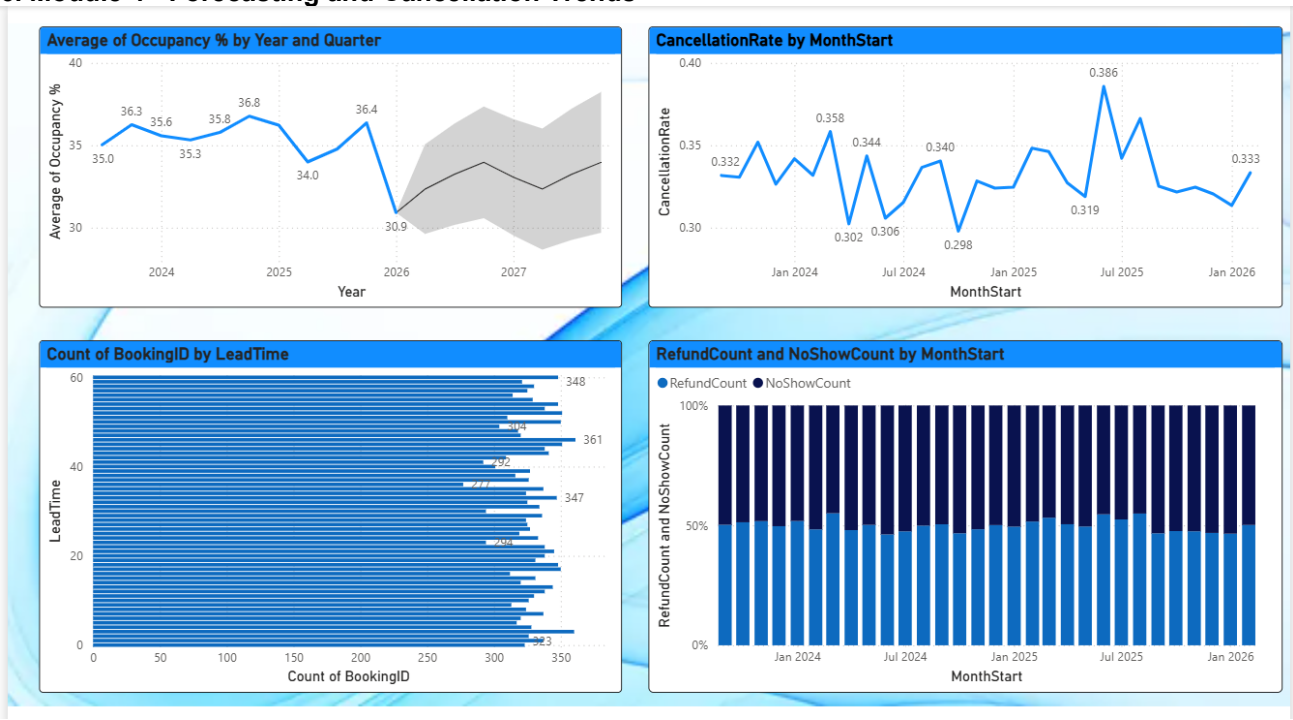
5. Module 3 - Guest Analysis Module

Data
Queries
Insert
Calculations
Sensitivity
Share
Copilot
Count of BookingID by Nationality and GuestType
GuestType Business Corporate Family Solo
Nationality
Indian 155 203 134 238
Australian 143 205 128 243
German 160 211 106 238
Canadian 132 207 111 255
French 154 192 108 235
Japanese 149 190 115 235
American 126 187 125 223
Brazilian 128 166 103 245
British 123 169 84 237
Chinese 101 168 110 191
0
500
Count of BookingID
Count of BookingID by BookingChannel and GuestType
GuestType Business Corporate Family Solo
100%
50%
0%
Count of BookingID
34.38% 34.77% 33.13% 36.76%
16.90% 16.67% 17.17% 16.02%
29.18% 28.42% 27.41% 27.69%
19.53% 20.14% 22.29% 19.53%
Travel Agent
Website
Call Center
Mobile App
BookingChannel
Average of Duration by GuestType
4.03 4.03 4.00 3.90
Average of Duration
Family
Solo
Corporate
Business
GuestType
Count of BookingID by BookingStatus
2.23K (33.14%)
2.27K (33.64%)
2.24K (33.22%)
BookingStatus
Checked-in
No-show
Cancelled
Gender
Female
Male
Other
Count of BookingID by PaymentMethod
1.66K (24.67%)
1.7K (25.31%)
1.67K (24.8%)
1.7K (25.22%)
PaymentMethod
UPI
Cash
Credit Card
Corporate Ac...
Sum of Revenue by PaymentMethod
₹62M (24.41%)
₹65M (25.76%)
₹63M (24.81%)
₹63M (25.02%)
PaymentMet...
UPI
Credit Card
Cash
Corporate ...

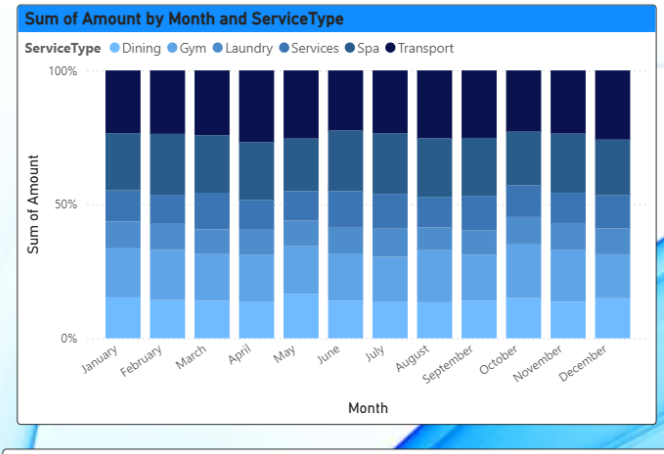

CustomerCluster, GuestType, Sour...
First-Timer
High Spender
Loyal Guest
Other
Count of SpendScore and Sum of FrequencyScore by RecencyScore and CustomerCluster
CustomerCluster First-Timer High Spender Loyal Guest Other
Sum of FrequencyScore
800
600
400
200
0
0
5
10
15
20
25
30
35
Count of SpendScore

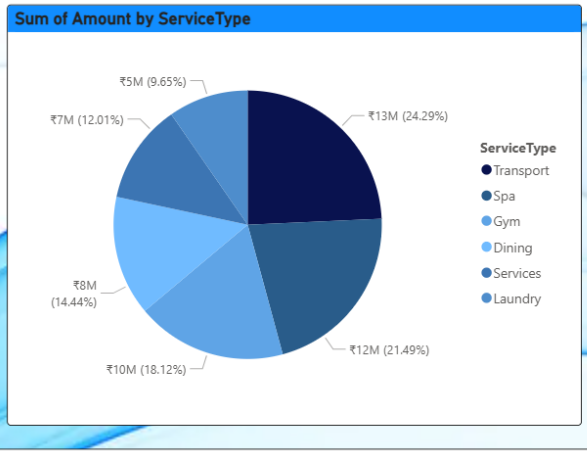


6. Module 4 - Forecasting and Cancellation Trends



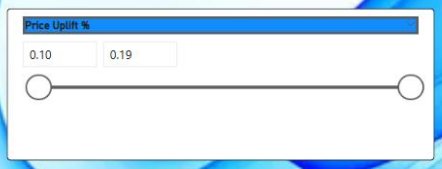
7. Module 5 - Revenue Strategy Dashboard

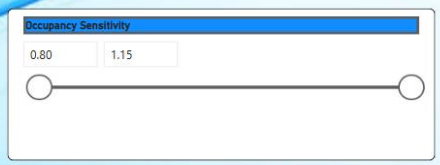




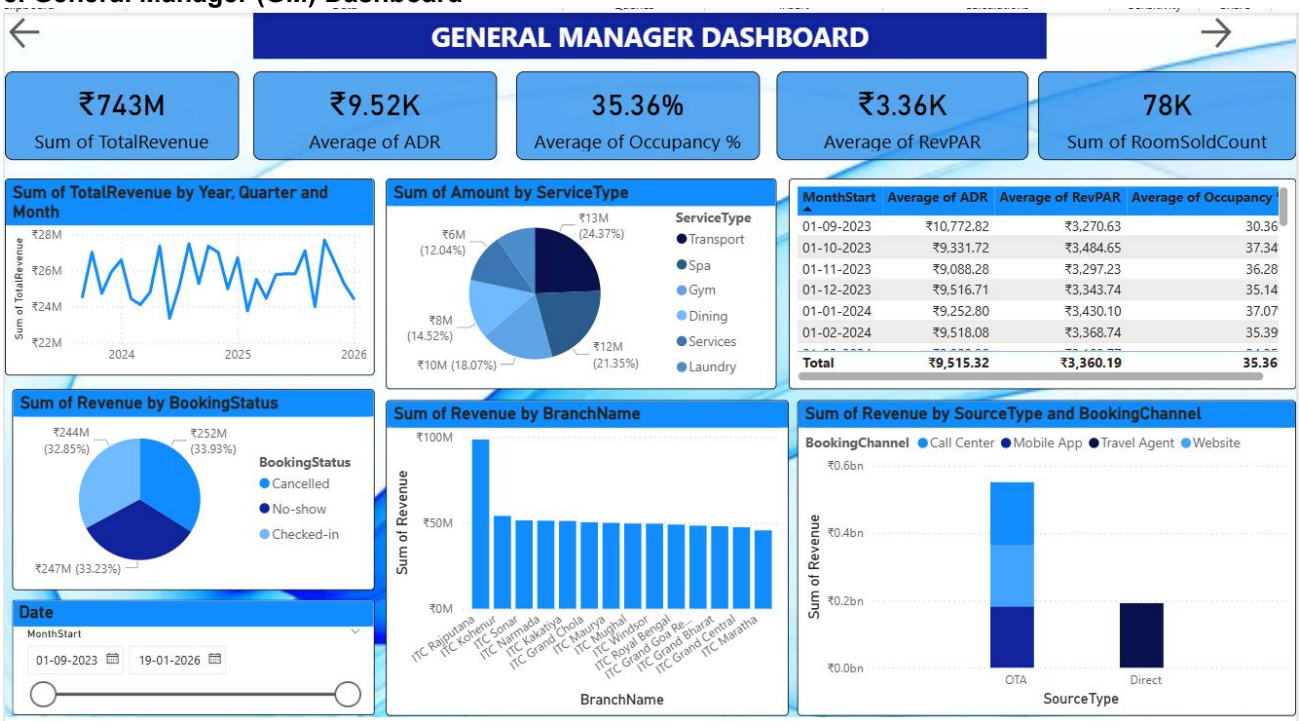
Year	2023		2024		2025		2026		Total	
ServiceType	Sum of Amount	Count of TransID	Sum of Amount	Count of TransID	Sum of Amount	Count of TransID	Sum of Amount	Count of TransID	Sum of Amount	Count of TransID
Dining	₹9,73,200	400	₹34,39,200	1427	₹31,38,000	1330	₹4,18,800	177	**₹79,69,200**	**3334**
Gym	₹13,35,000	446	₹41,05,500	1368	₹39,55,500	1313	₹6,04,500	206	**₹1,00,00,500**	**3333**
Laundry	₹7,50,400	467	₹20,92,000	1317	₹22,01,600	1372	₹2,80,800	177	**₹53,24,800**	**3333**
Services	₹8,66,000	438	₹27,00,000	1364	₹26,91,000	1346	₹3,71,000	186	**₹66,28,000**	**3334**
Spa	₹15,62,400	431	₹47,61,000	1360	₹48,04,200	1341	₹7,34,400	201	**₹1,18,62,000**	**3333**
Transport	₹18,06,000	443	₹54,58,000	1359	₹54,42,000	1359	₹7,00,000	172	**₹1,34,06,000**	**3333**
Total	**₹72,93,000**	**2625**	**₹2,25,55,700**	**8195**	**₹2,22,32,300**	**8061**	**₹31,09,500**	**1119**	**₹5,51,90,500**	**20000**

RoomType	High	Low	Shoulder	Total
Conference Room	11,500.00	9,000.00	10,500.00	**10,000.00**
Deluxe	6,325.00	4,950.00	5,775.00	**5,500.00**
Dormitory	1,380.00	1,080.00	1,260.00	**1,200.00**
Double	4,600.00	3,600.00	4,200.00	**4,000.00**
Economy	2,070.00	1,620.00	1,890.00	**1,800.00**
Executive Suite	13,800.00	10,800.00	12,600.00	**12,000.00**
Family Suite	10,925.00	8,550.00	9,975.00	**9,500.00**
Luxury Villa	34,500.00	27,000.00	31,500.00	**30,000.00**
Penthouse	28,750.00	22,500.00	26,250.00	**25,000.00**
Presidential Suite	23,000.00	18,000.00	21,000.00	**20,000.00**
Single	2,875.00	2,250.00	2,625.00	**2,500.00**
Studio	5,175.00	4,050.00	4,725.00	**4,500.00**
Suite	9,200.00	7,200.00	8,400.00	**8,000.00**
Superior	6,900.00	5,400.00	6,300.00	**6,000.00**
Twin	4,370.00	3,420.00	3,990.00	**3,800.00**
Total	**10,715.98**	**8,494.49**	**10,222.33**	**9,493.40**

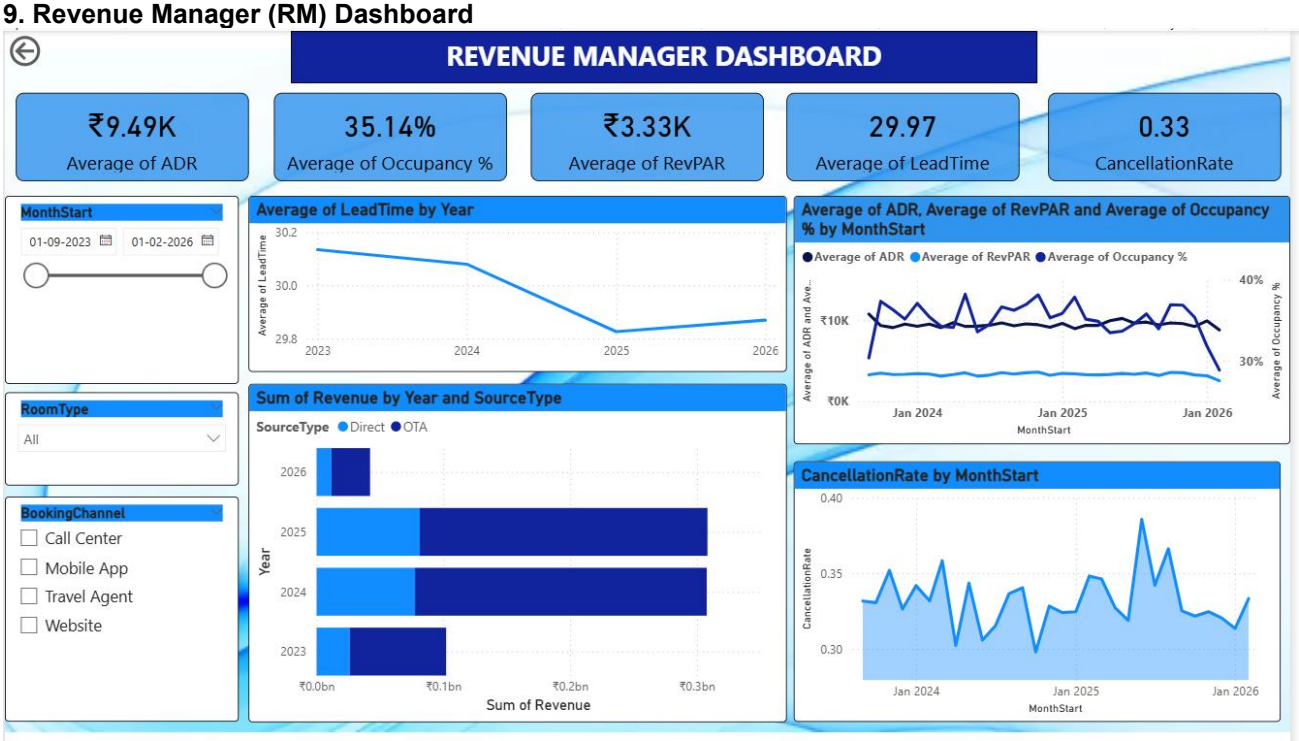




8. General Manager (GM) Dashboard



9. Revenue Manager (RM) Dashboard



7. Challenges Face

During the development and implementation of the **HotelRevAI – AI-Driven Revenue Analysis for Hotels** project, several technical, analytical, and design-related challenges were encountered. These challenges were systematically addressed to ensure the successful completion of the project.

- **Data Quality and Consistency Issues**
  Hotel booking datasets contained missing values, inconsistent date formats, and varied representations of booking channels and room categories. Significant effort was required to clean, standardize, and validate the data before analysis.
- **Complexity in Data Modeling**
  Designing a scalable **star schema** that accurately represented booking facts while supporting multiple dimensions such as date, room type, customer, and hotel branch required careful planning to avoid redundancy and ensure efficient querying.
- **Accurate KPI Calculation**
  Metrics like Occupancy %, ADR, and RevPAR require precise business logic. Ensuring correct calculations across different time granularities (daily, weekly, seasonal) while avoiding aggregation errors was a major analytical challenge.
- **Guest Segmentation and Classification**
  Segmenting guests into categories such as business, family, loyal guests, and high spenders involved selecting meaningful features and validating segmentation logic to ensure business relevance.
- **Forecasting and Trend Interpretation**
  Predicting future occupancy and interpreting cancellation trends required careful handling of historical data patterns and seasonality to avoid misleading forecasts.
- **Dashboard Design and Usability**
  Creating a single dashboard that balances detailed analytics with clarity for both **General Managers** and **Revenue Managers** was challenging. The dashboard needed to remain interactive, intuitive, and decision-focused without overwhelming users.

8. Learnings & Skills Acquired

The **HotelRevAI** project provided extensive hands-on exposure to hospitality analytics, data modeling, and business intelligence tools. The key learnings and skills acquired during the project include:

- **Hospitality Domain Knowledge**
  Gained a strong understanding of hotel revenue management concepts such as occupancy rate, ADR, RevPAR, booking lead time, and cancellation behavior.
- **Data Modeling & ETL Skills**
  Learned to design and implement a star schema, handle fact–dimension relationships, and create

calculated fields for analytical reporting.

- **Data Analysis & KPI Development**
  Developed expertise in calculating and validating business KPIs and analyzing trends across time, customer segments, and booking channels.
- **Guest Segmentation & Behavioral Analysis**
  Acquired skills in customer profiling, segmentation, and clustering to identify loyal guests, high-value customers, and different traveller types.
- **Forecasting & Trend Analysis**
  Learned to analyze historical booking data to identify demand patterns, seasonal trends, and future occupancy forecasts.
- **Dashboard Development & Storytelling**
  Improved proficiency in building interactive dashboards using Power BI, applying slicers, drill-downs, and visual storytelling techniques to present insights effectively.
- **Analytical Thinking & Decision Support**
  Strengthened the ability to translate raw data into actionable insights that support strategic decision-making for hotel management.

9. Testimonials from Team

The **HotelRevAI** project team demonstrated strong analytical capabilities, consistent collaboration, and a clear understanding of business-driven data analytics. The project was executed with a structured approach, meeting all planned milestones within the given timeline.
The team showed commendable dedication in understanding hospitality revenue concepts, implementing accurate KPIs, and designing an intuitive dashboard tailored for hotel decision-makers. Their ability to combine technical skills with business insights reflects a high level of professionalism and readiness for real-world analytics challenges.

10. Conclusion

The **HotelRevAI – AI-Driven Revenue Analysis for Hotels** project successfully demonstrates the application of data analytics and AI concepts in the hospitality domain. By integrating data modeling, performance metrics, guest analysis, and forecasting into a unified dashboard, the project delivers actionable insights that support strategic revenue management.

The project enhanced practical understanding of hotel KPIs, data visualization, analytical thinking, and business-oriented storytelling. Overall, HotelRevAI serves as a strong foundation for advanced analytics-driven decision support systems in the hospitality industry.

11. Acknowledgements

I sincerely thank the mentors, coordinators, and reviewers for their valuable guidance and continuous support throughout the project. Their feedback and direction played a crucial role in the successful completion of this project. I am also grateful for the opportunity to apply analytical and AI concepts to a real-world hospitality use case.